x0xb0x Mainboard Beta Release (3/8/2005)

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CPU

File: p02_cpu.kicad_sch

LEDs

File: p05_leds.kicad_sch

VCA

File: p08_vca.kicad_sch

Rotary encoders

File: p03_encoders.kicad_sch

CV and VCO

File: p06_cv_vco.kicad_sch

Envelope

File: p09_env.kicad_sch

Tact switches

File: p04_switches.kicad_sch

VCF

File: p07_vcf.kicad_sch

Mixer and headphone

File: p10_mixer.kicad_sch

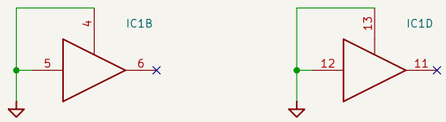
x0xb0x mainboard

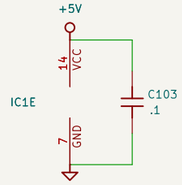
Sheet: /
File: mainboard.kicad_sch

Title: x0xb0x mainboard

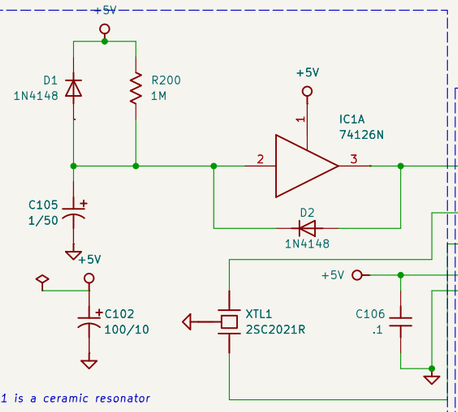
Size: A3	Date:	**Rev: K29**
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Unused gates of IC1



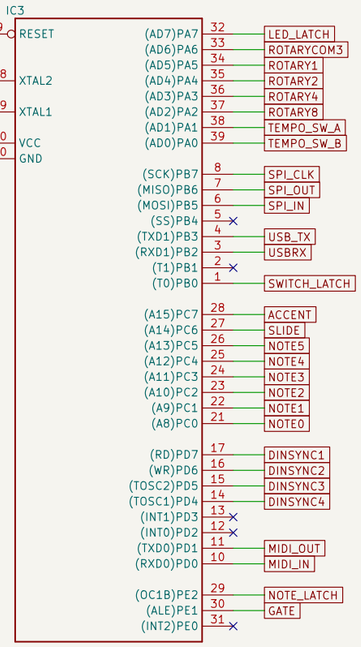


Reset and clock

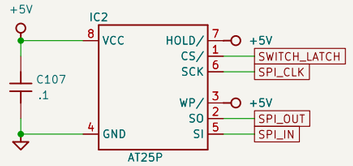


XTL1 is a ceramic resonator

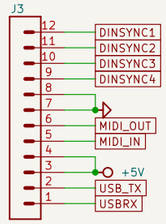
MCU



EEPROM



To the I/O board



Mounting holes



3.3 mm holes: on the board only, not on the original sheet

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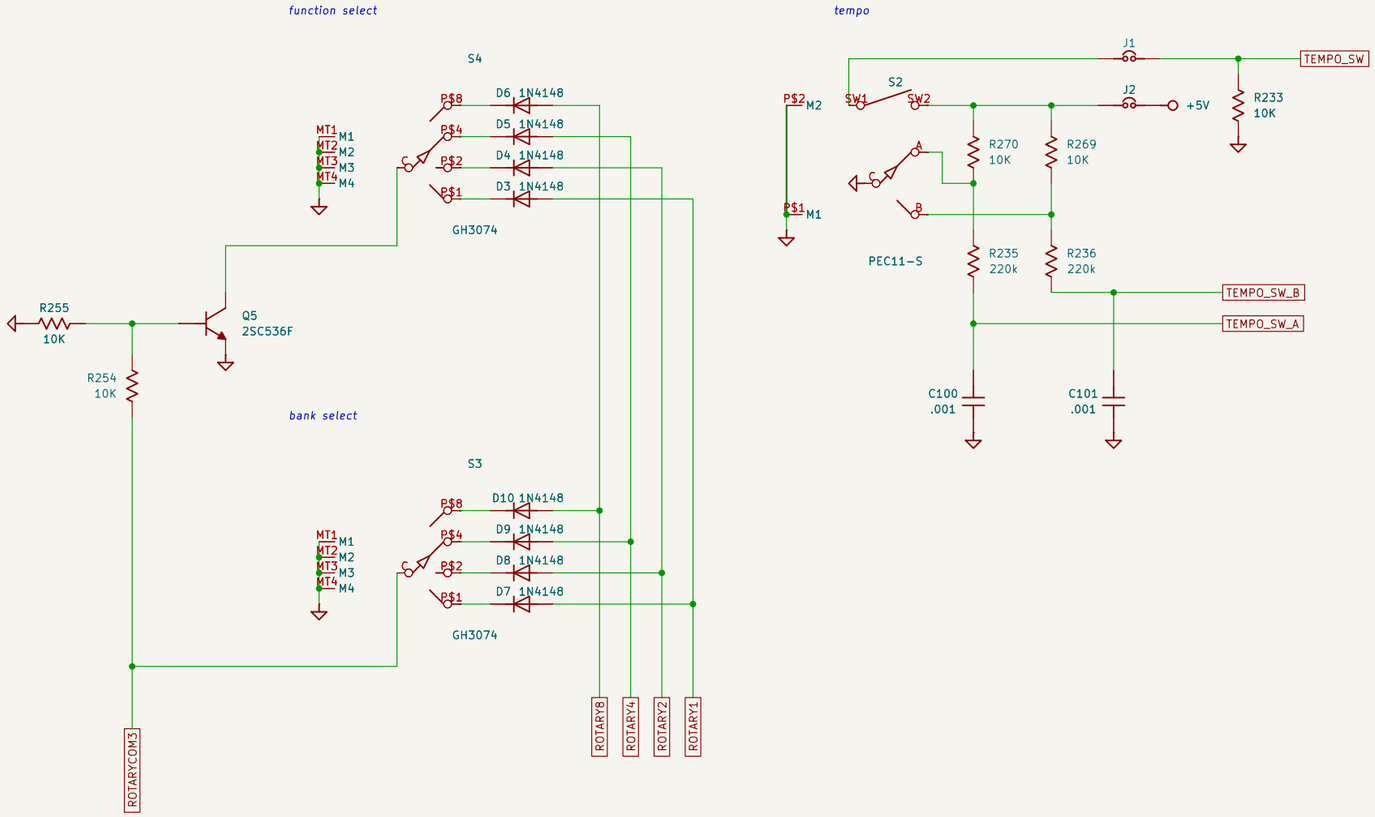
x0xb0x mainboard

Sheet: /CPU/
File: p02_cpu.kicad_sch

Title: CPU

Size: A3	Date:	**Rev: K29**
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Rotary Encoders



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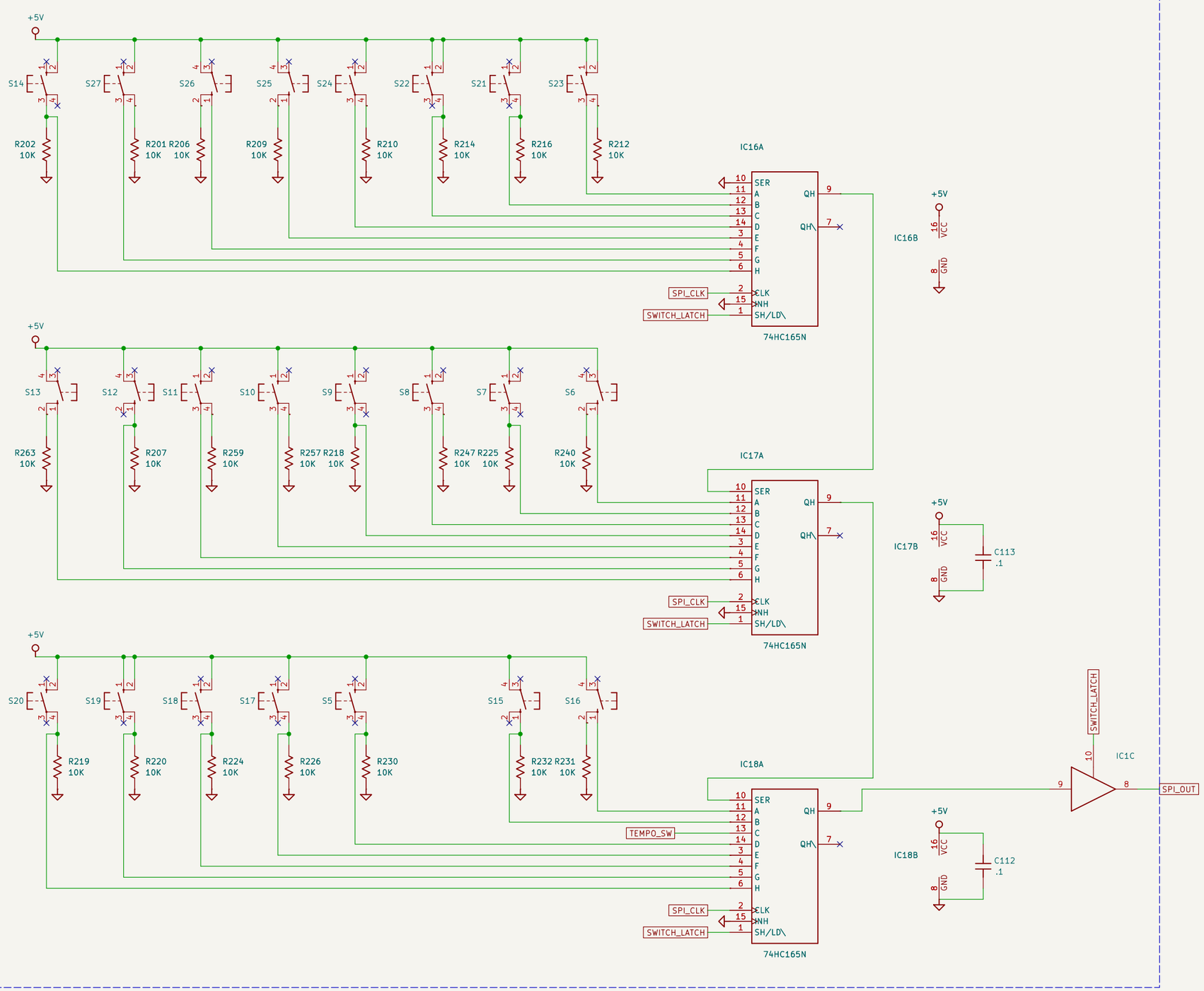
x0xb0x mainboard

Sheet: /Rotary encoders/
File: p03_encoders.kicad_sch

Title: Rotary encoders

Size: A3	Date:	**Rev: K29**
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Tact Switches



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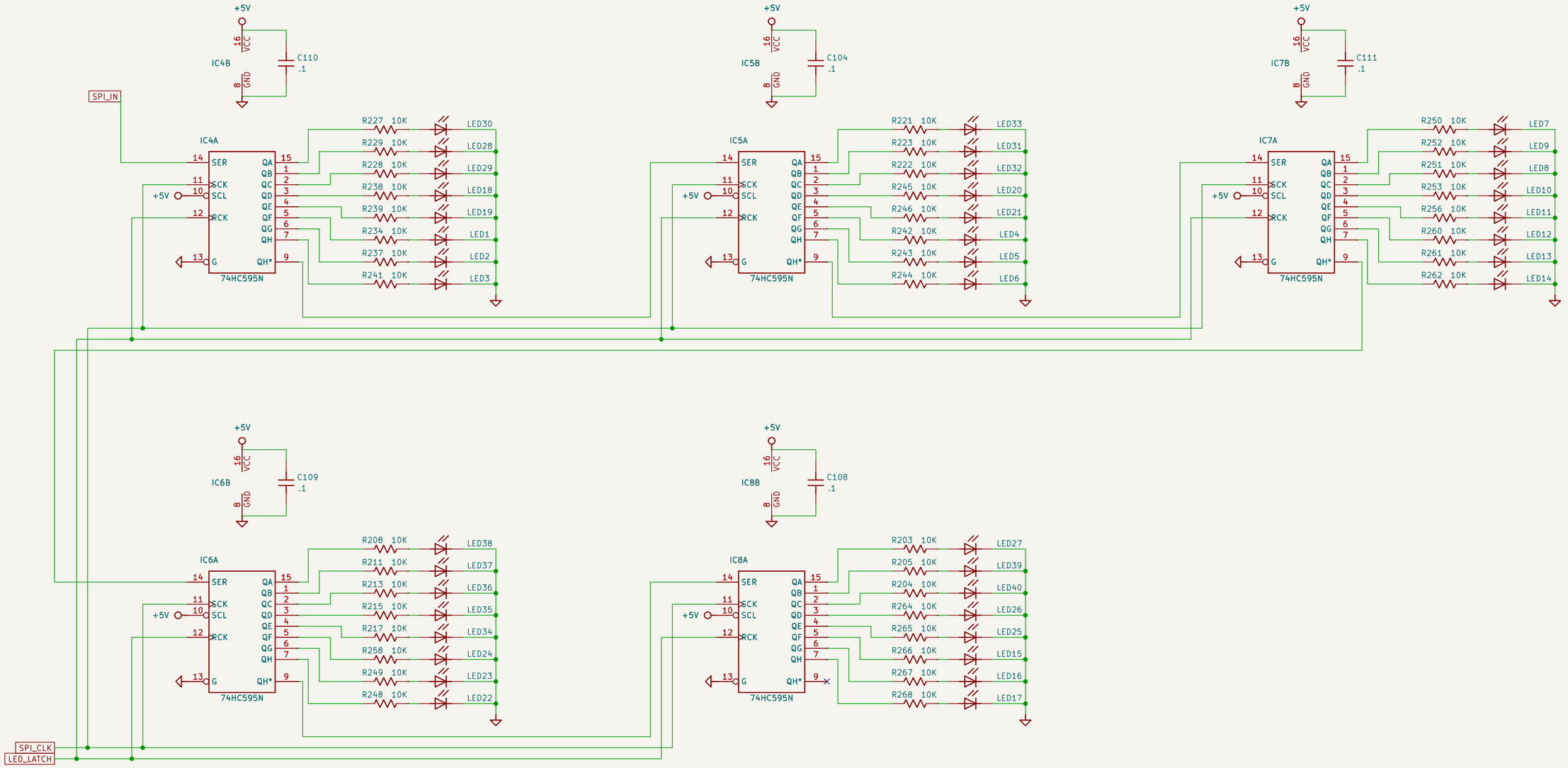
x0xb0x mainboard

Sheet: /Tact switches/
File: p04_switches.kicad_sch

Title: Tact switches

Size: A3	Date:	**Rev: K29**
KiCad E.D.A. 10.0.7		Id: 4/10

LEDs



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x0xb0x mainboard

Sheet: /LEDs/
File: p05_leds.kicad_sch

Title: LEDs

Size: A3	Date:	**Rev: K29**
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IC11C
+12V
+5.333V
IC9B
IC9A
4174P
NOTE0
NOTE2
NOTE4
NOTE5
NOTE3
NOTE1
NOTE_LATCH
R90 200k
R87 200K
R79 200K
R80 200K
R81 200K
R82 200K
R88 200K
R89 200K
R85 200K
R86 200K
R84 200K
R83 200K
R76 200K
R77 200K
R74 200K
R75 200K
R78 200K
IC11B
R107 2.2K
D31 1N4148
D30 1N4148
IC12C
4066N
IC12D
R93 100K
C35 .22
Q30 2SC536F
R91 1M
R59 100K
SLIDE
Q29 2SC536F
R60 22K
R92 100K
CV and gate to the I/O board
J6
N$622
TM4 50K(B)
R103 220k
R102 100K
TM5 5K(B)
R106 27K
R118 100K
VR2 50K(B)
TUNING
Q26 2SC1583
R100B0
R100A0
R104 2.2K
R100A0, R100B0: THERMISTOR-7,5
IC11A
C34 .001
Q27 2SA733P
Q25 2SC536F
D25 1N4148
Q24 2SC536F
R101 10K
Q28 2SK30
C33 .01
R105 10K
R45 22K
C11 1/50
Q8 2SA733P
R35 100K
R34 10K
C10 .01
R36 10K
WAVEFORM
S1
N$126
supply from the I/O board
C32 10/16
C31 10/16
J4
SMD pair for Q26
Q126 BCM847BS
N$113
N$98
N$99
N$101
Fit Q126 or Q26, never both

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x0xb0x mainboard

Sheet: /CV and VCO/
File: p06_cv_vco.kicad_sch

Title: CV and VCO

Size: A3	Date:	**Rev: K29**
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Diode ladder

R115 10K
Q22 2SC2291
C30 10/16
C26 .033
Q16 2SC536F
Q23 2SC536F
C24 .033
Q14 2SC536F
Q17 2SC536F
C19 .033
Q13 2SC536F
Q15 2SC536F
C18 .018
Q12 2SC1583
from the waveform switch S1
N$126
R62 220K
C17 1/50
R70 2.2K
R98 2.2K
+5.333V
R68 2.2K
C16 10/16
R69 2.2K

Output amplifier and resonance

R95 100
+12V
R111 22K
R110 22K
C28 47/16
Q19 2SC536FUNEVEN
C14 1/50
VR4A
N$27
to C21 and the VCA
N$26
to C20 and the VCA
R96 10K
50K(B)
C25 .1
C27 .1
Q21 2SC1583
R113 100K
R114 100K
+5.333V
R116 10K
RESONANCE
R112 10K
C29 1/50
Q20 2SC536F
R109 10K
R108 2.2K
Q18 2SC536F
R67 2.2K
C15 1/50
R99 100K
R97 10K
C23 1/50
R66 100K
+5.333V
R94 10K
C22 1/50

SMD pairs

Q122 BCM847BS
N$122
N$122
N$143
N$115
N$120

Q112 BCM847BS
N$118
N$121
N$124
N$60
N$127

Q121 BC847CDW1T1G
N$140
N$148
N$138
N$141
N$139

Fit each chip or the SIP part it replaces, never both

Exponential converter

Q11 2SC536F
R73 100K
TM3 500K(B)
VR5
N$53
from MEG
VR3
+12V
50K(A)
Q10 2SA733P
R63 220K
50K(A)
CUTOFF FREQ.
R71 2.2K
R72 100K
ENV. MOD
+5.333V
R65 10K
R61 10K
Q9 2SA733P
R47 10K
R64 10K

Accent sweep

VR4B
50K(B)
C13 1/50
R46 47K
D24 1N4148
N$35
accent line, from R119 and VR7

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x0xb0x mainboard

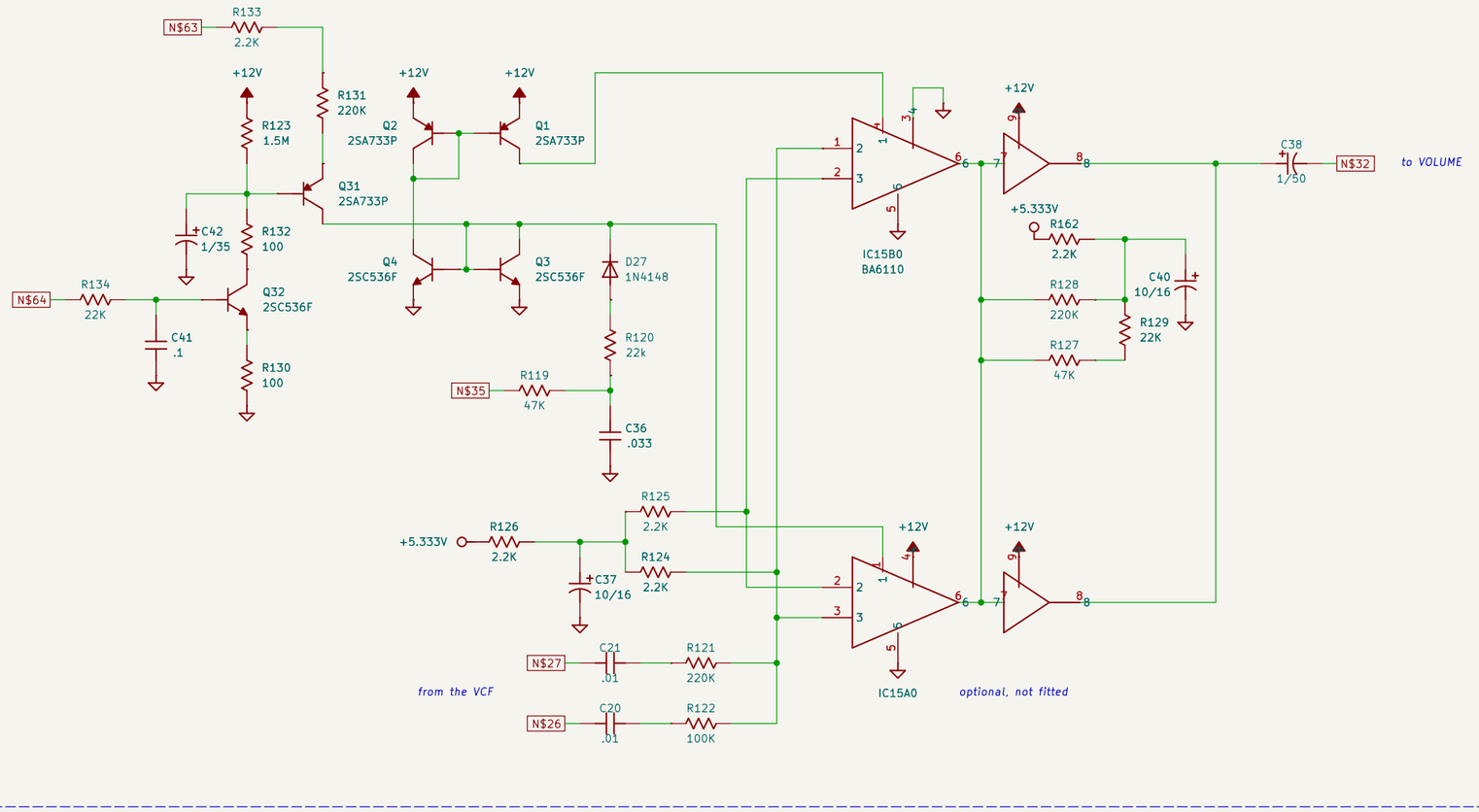
Sheet: /VCF/
File: p07_vcf.kicad_sch

Title: VCF

Size: A3	Date:	**Rev: K29**
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VCA

the envelope, turned into a current



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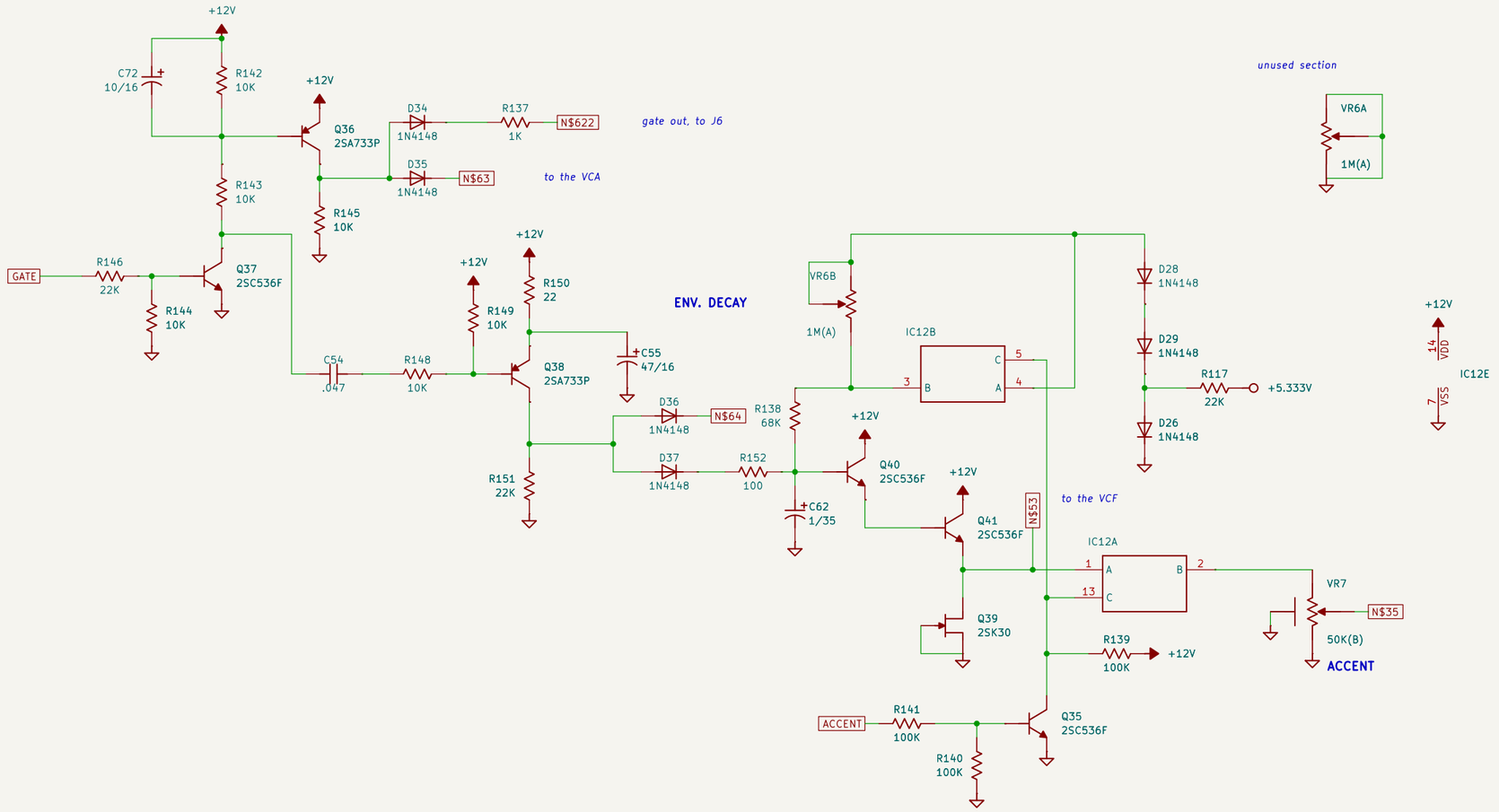
x0xb0x mainboard

Sheet: /VCA/
File: p08_vca.kicad_sch

Title: VCA

Size: A3	Date:	**Rev: K29**
KiCad E.D.A. 10.0.7		Id: 8/10

ENV



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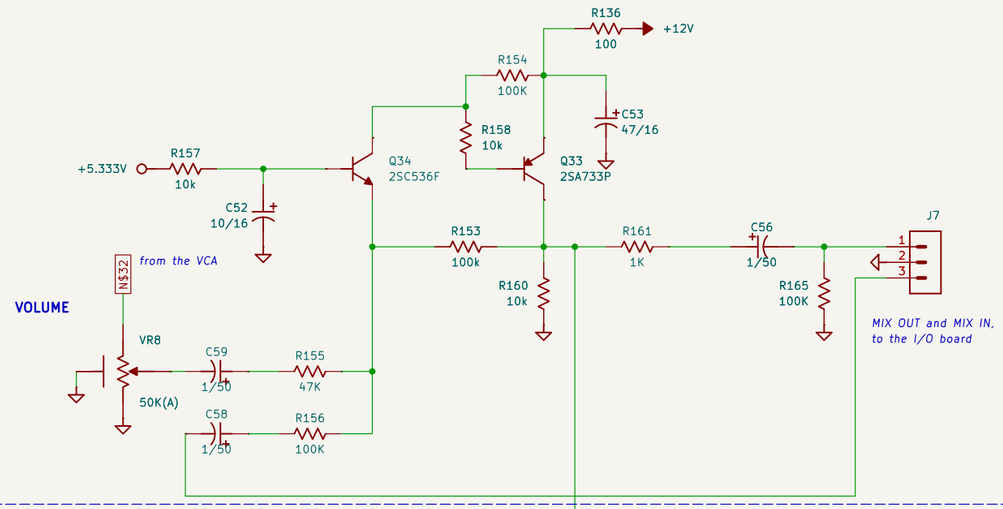
x0xb0x mainboard

Sheet: /Envelope/
File: p09_env.kicad_sch

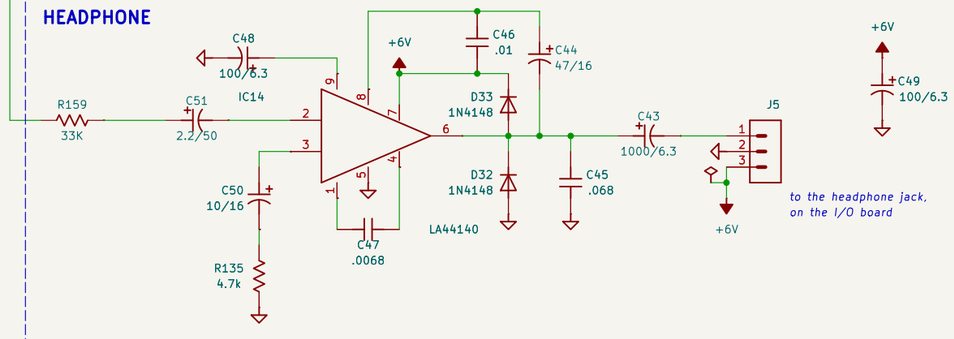
Title: Envelope

Size: A3	Date:	**Rev: K29**
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MIXER



HEADPHONE



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x0xb0x mainboard

Sheet: /Mixer and headphone/
File: p10_mixer.kicad_sch

Title: Mixer and headphone

Size: A3	Date:	**Rev: K29**
KiCad E.D.A. 10.0.7		Id: 10/10